

Atomic Spectroscopy

Outline

❖ Introduction

❖ Types of atomic spectroscopy

❖ Instrumentation

❖ Applications

Introduction

❖ Atomic Spectroscopy - interaction of an **atom** in the **gas phase** with EMR.

- Based upon the ability of atoms to absorb or emit light.
- Sample is atomized (atoms)
- Absorption or emission measured

Types of Atomic Spectroscopy

❖ The absorption or emission of light by vapor-state atoms may be measured.

➤ Atomic absorption spectroscopy

➤ Atomic emission spectroscopy

Atomic Absorption Spectroscopy

- ❖ Measure concentration by absorption of light
- ❖ Requires a light source of proper wavelength

Instrumentation (AAS)

➤ AAS: Single-beam or Double-beam

1. Radiation source
2. Atomizer
3. Monochromator
4. Detector & read-out devices

1. Radiation source

Hollow cathode lamp (HCL)

- ❖ Light source used in an AAS
- ❖ Contains a coated cathode of the element that is to be analysed.
- ❖ Emits a beam of a specific wavelength across the burner and into the monochromator.

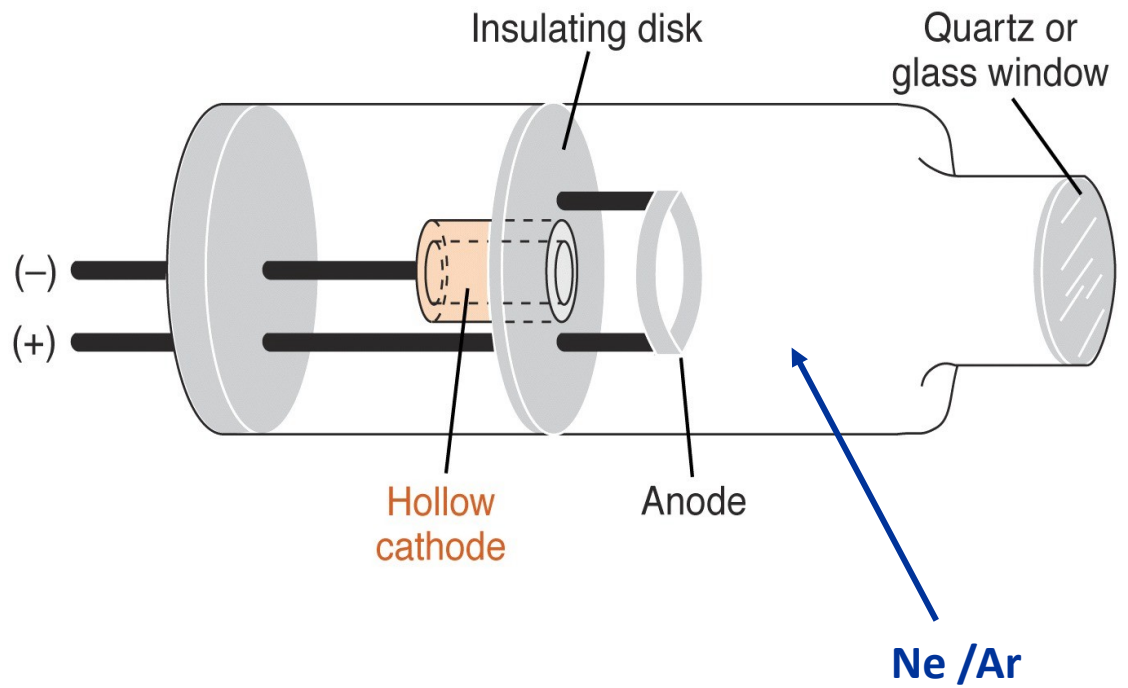
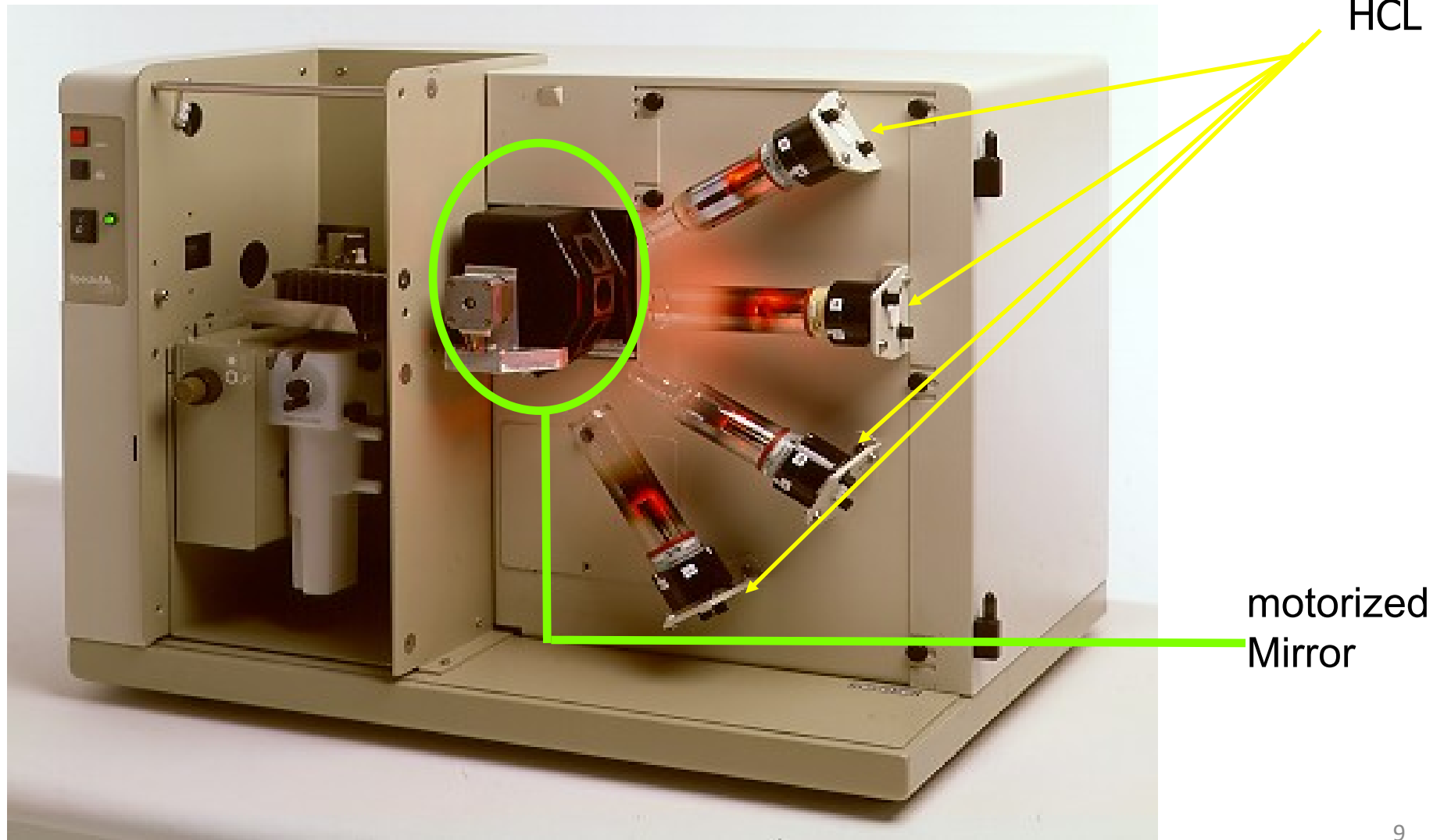


Fig. Hollow cathode lamp

AAS



HCL...

❖ Both anode & cathode are sealed in

➤ Glass cylinder

- Filled with the carrier gas of Argon or Neon.
- At its end is a window transparent to the emitted radiation.

❖ A lamp filled with

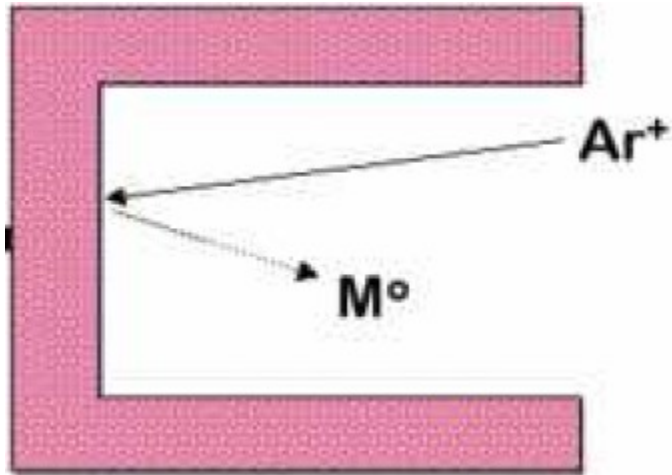
➤ Neon will produce a red beam.

➤ Argon will produce a blue beam.

HCL emission process

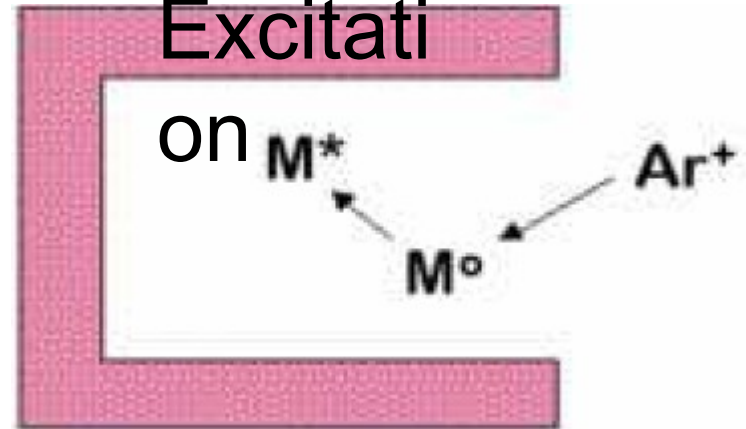
- ❖ When an electrical potential is applied between the anode & cathode, some of the fill gas atoms are ionized.
- ❖ The +vely charged fill gas ions accelerate through the electrical field to collide with the -vely charged cathode
 - Dislodge individual metal atoms in a process called “sputtering”.
- ❖ Sputtered metal atoms are then excited to an emission state through a kinetic energy transfer by impact with fill gas ions.
 - Emit photons

1. Sputtering



2.

Excitation



3.

Emission

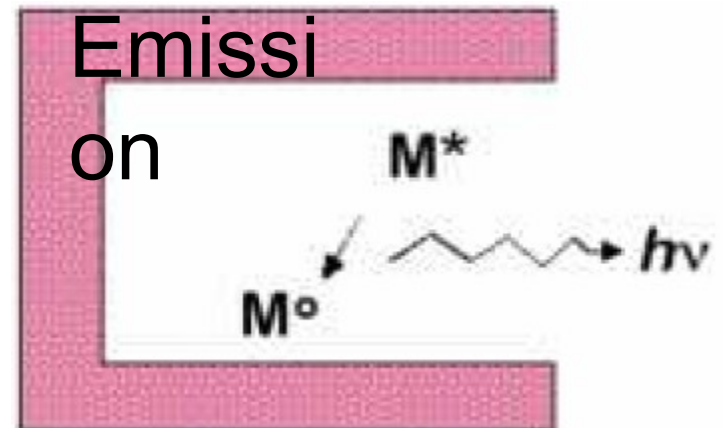


Fig. Hollow cathode lamp process

2. Atomizers

❖ Atomization

➤ Process of breaking analyte into gaseous atoms

1. Flame

2. Furnace (graphite furnace)

3. Plasma (Inductively Coupled Plasma)

Flame Atomization

- ❖ The sample is first converted into a fine mist consisting of small droplets of solution by nebulizer.
- ❖ The sample is aspirated into a spray chamber by passing a high-pressure stream consisting of one or more combustion gases.
- ❖ The impact of the sample with the glass impact bead produces an aerosol mist.

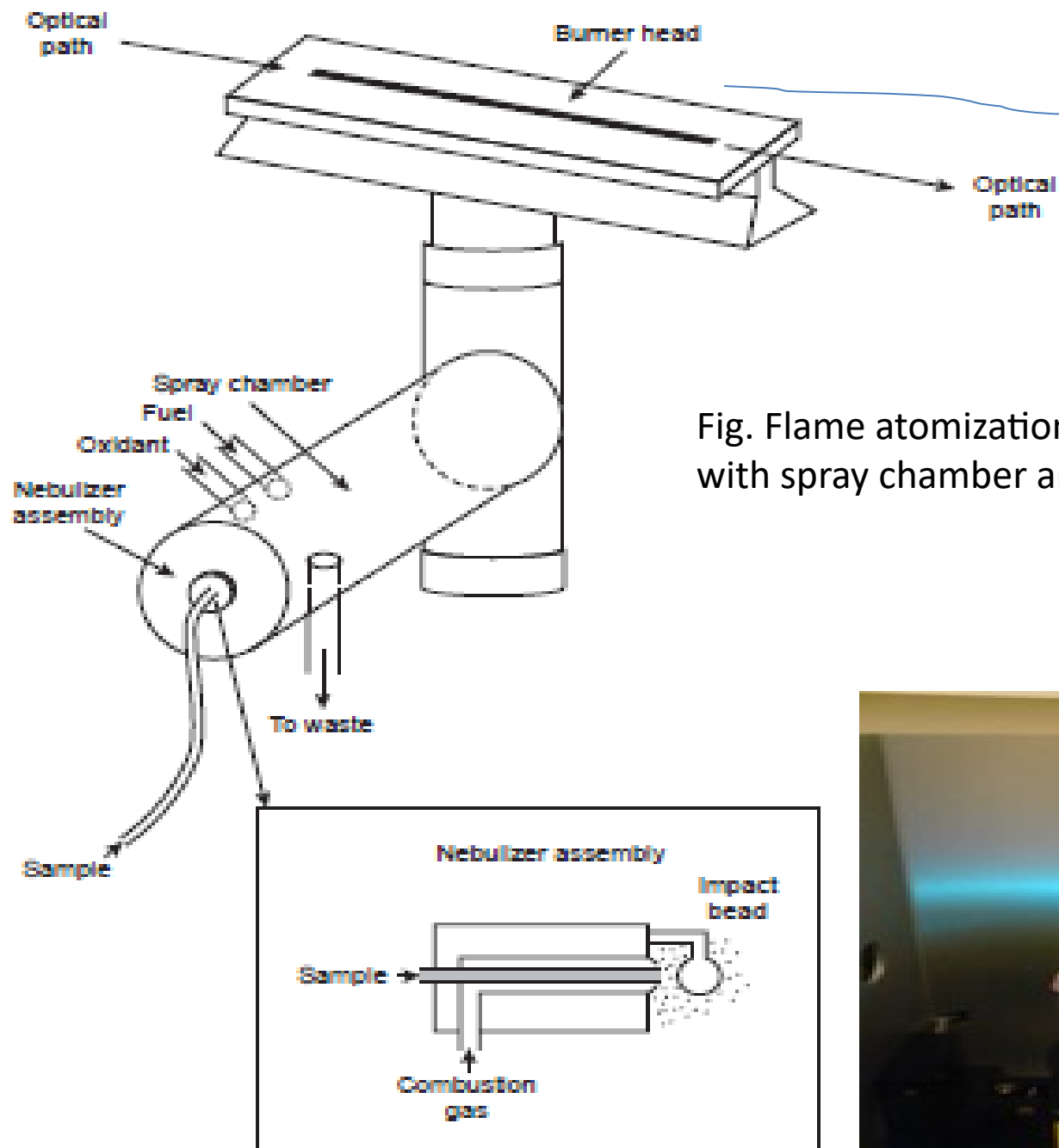
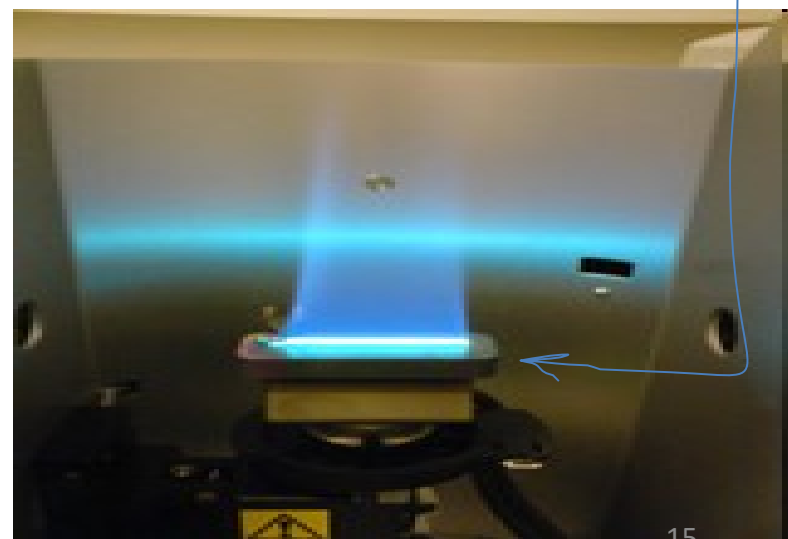


Fig. Flame atomization assembly equipped with spray chamber and slot burner.



Processes occur during atomization:

- a. Nebulized samples are sprayed into a flame as a spray of very fine droplets
- b. Droplets will lose their solvent content due to very high flame temperatures in a process called **desolvation**
 - And will thus be converted into a solid aerosol.
- c. The solid aerosol is **volatilized** to form gaseous molecules

Processes occur during atomization...

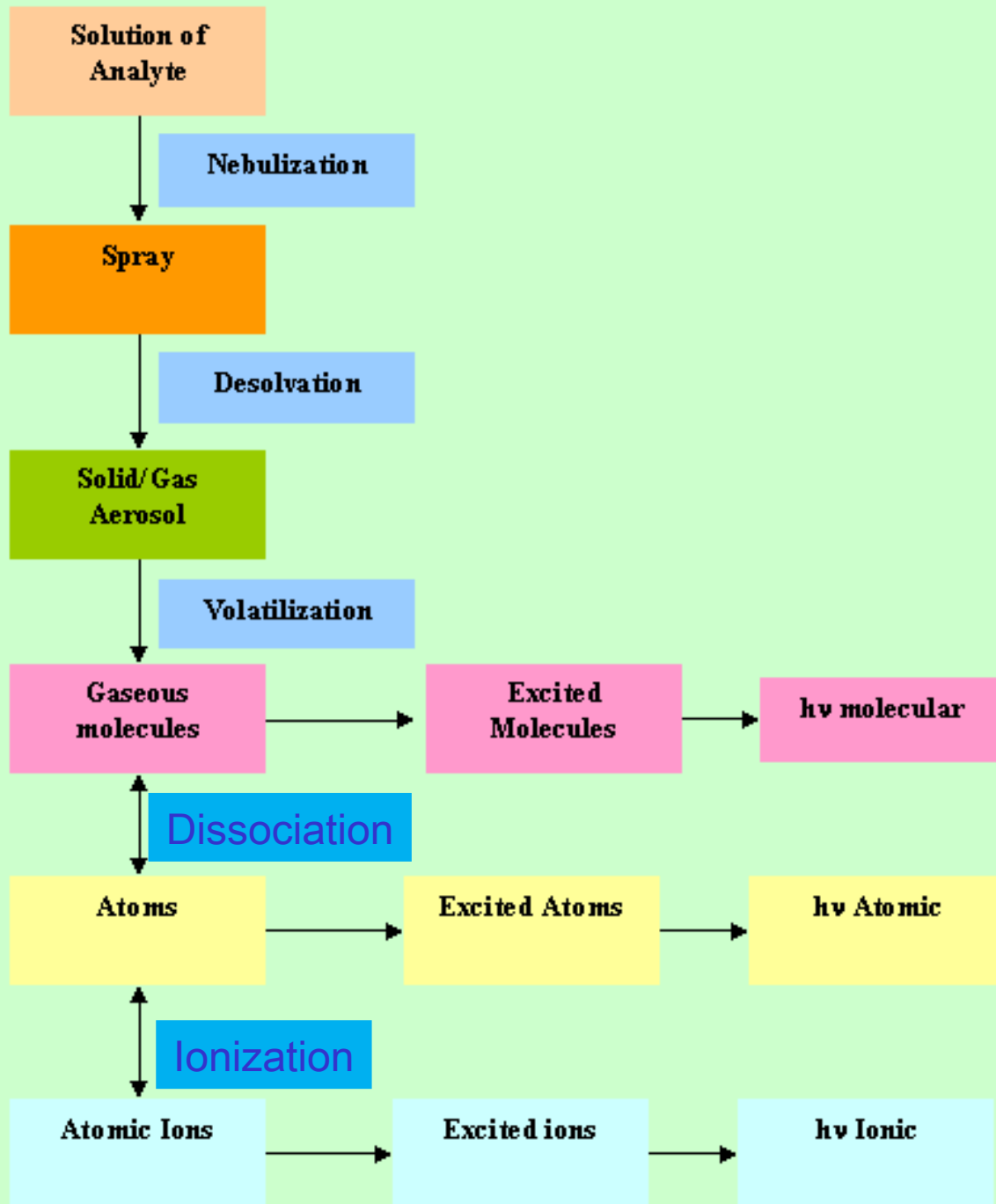
d. Gaseous molecules will then be **atomized** and neutral atoms are obtained.

➤ If **energy is not enough for atomization**, gaseous molecules will not be atomized

- Molecular absorption or emission

e. Atoms in the gaseous state can absorb energy and are excited.

➤ If **energy is too much**, we may observe **ionization**.



The flame breaks down the analyte's matrix
→ create the elemental form of the analyte atom

Scheme: the different processes occurring during atomization

❖ Thermal energy in flame atomization is provided by the combustion of a **fuel–oxidant** mixture.

❖ Common oxidants/fuels:

➤ Air–acetylene → for easily ionised elements

➤ Nitrous oxide–acetylene → for more difficult elements

❖ The slot burner provides a long path length for monitoring absorbance and a stable flame.

The flame can be **adjusted** by altering the **gas flow rate** to produce three different types of flame conditions.

Oxidising flame

- ❖ Very weak flame /observed to be blue.
- ❖ Used for elements that are easily atomised like cadmium, lead, nickel etc..

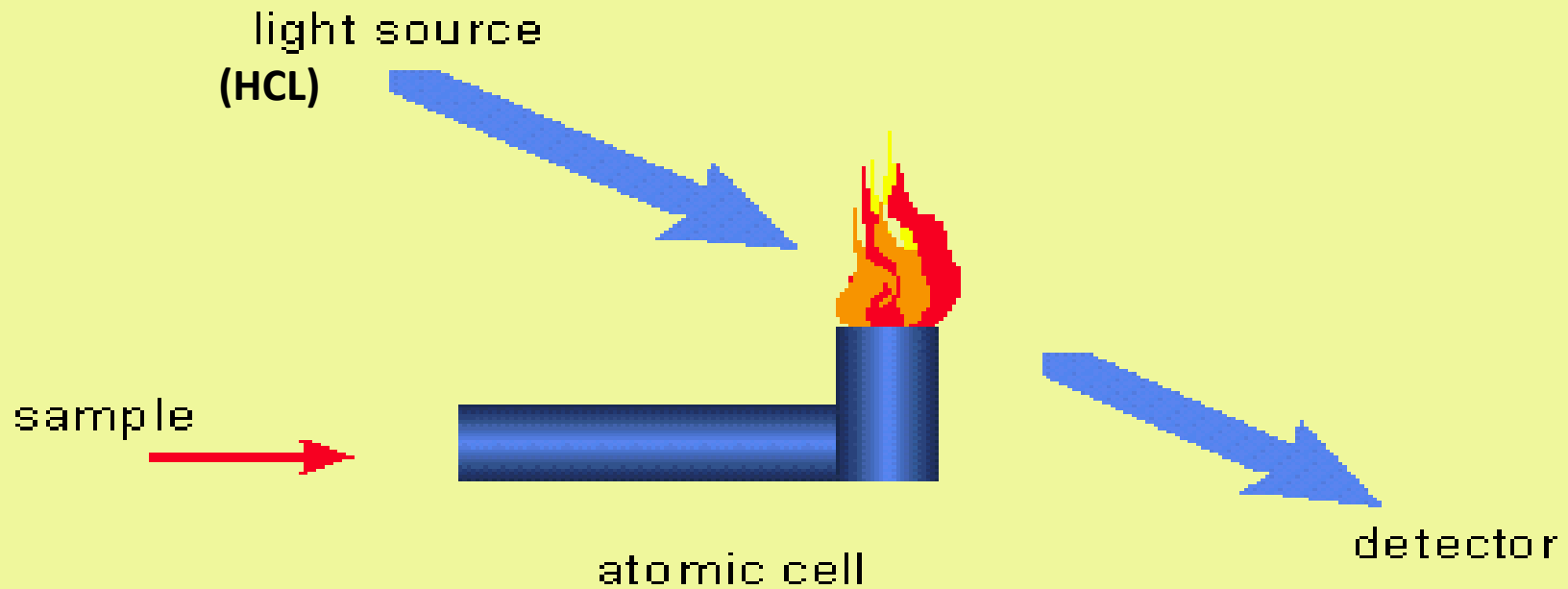
Reducing flame

- ❖ A fuel rich flame which produces an excess of Carbon and Hydrogen atoms
 - Help to breakdown the oxide bonds which form with some element like chromium, tin etc
 - Observed to be bright yellow/white.

Stoichiometric flame

- ❖ Falls in the middle of the above.
- ❖ Its appearance is blue with a yellow translucent band above.
- ❖ Used for elements that are not so easily atomised like iron etc.

Flame atomization



Merits of flame atomization

- ❖ Good reproducibility/good precision
- ❖ Virtually no spectral interference
- ❖ Ease of use
- ❖ Low initial cost
- ❖ Low running costs

Demerits

❖ The efficiency of atomization may be quite poor.

➤ This may occur for two reasons.

1. The majority of the aerosol mist produced during nebulization consists of droplets that are too large to be carried to the flame by the combustion gases.

✓ Consequently, as much as 95% of the sample never reaches the flame.

2. The large volume of combustion gases significantly dilutes the sample.

✓ Reduced sensitivity since the analyte's conc in the (flame <<< solution). Detection limit high only ppm

Electrothermal atomizer/Graphite furnace

- ❖ A tube of graphite is located in the sample compartment of the AAS, with the light path passing through it.
- ❖ Samples are injected into the graphite tube through a small hole located at the top of the tube.
- ❖ There is no nebulization.
- ❖ A power supply is used to pass a current through the graphite tube, resulting in resistive heating.

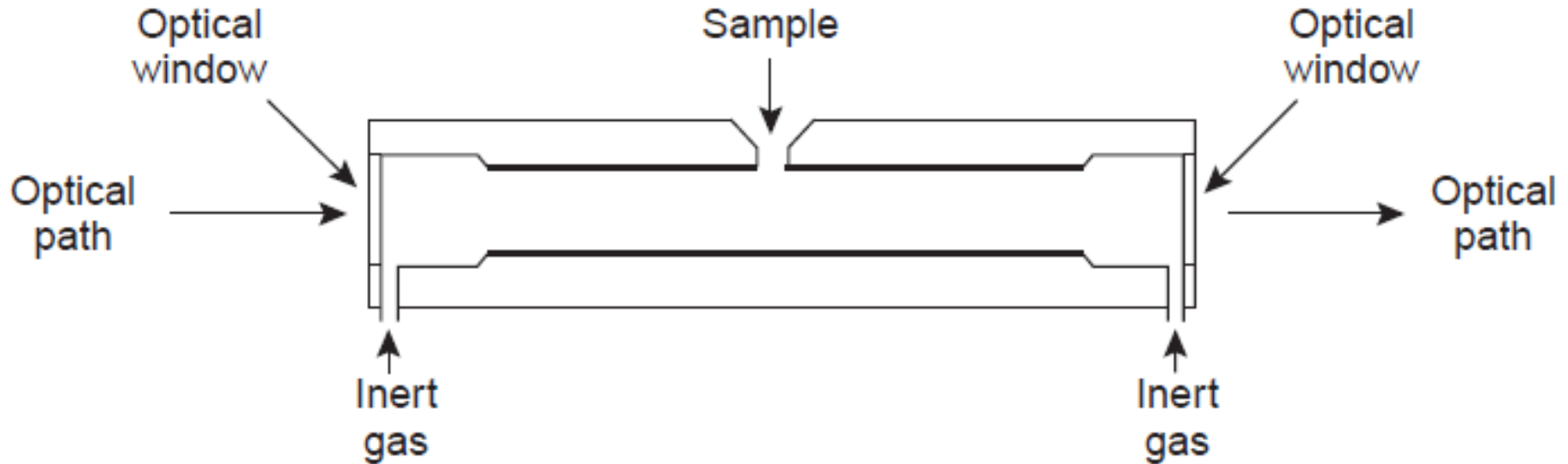


Fig. Electrothermal analyzer

❖ A continuous stream of inert gas:

- Protects the graphite tube from oxidation,
- Removes the gaseous products produced during atomization.

- ❖ The graphite is heated to a temp which is high enough to evaporate the solvent from the solution.
- ❖ The current is then increased so that
 - The sample is **ashed** & then ultimately it **vaporises** & **dissociates** into gaseous atoms.
- ❖ The light from the source (HCL)
 - Passes via the furnace & absorption during the atomization step is recorded over several seconds.
- ❖ ETAAS or GFAAS

Atomization is achieved in three stages

Drying

- $T^{\circ} \uparrow$ to about 110°C
- Evaporation of solvents & leaves the sample as a solid residue.

Ashing

- $T^{\circ} \uparrow$ to $350\text{--}1200^{\circ}\text{C}$
- Any organic material is converted to CO_2 & H_2O .
- Volatile inorganic materials are vaporized.

Atomiz'n

- $T^{\circ} \uparrow$ to $2000\text{--}3000^{\circ}\text{C}$ rapidly
- Atomize the analyte

Graphite furnace...

Merits

- Low detection limit
normally ppb
- Excellent sensitivity
- Low sample volume
- Direct analysis of solid samples
- No combustible gasses required

Demerits

- Poor precision
- Background absorption effects
- Analyte may be lost at the
ashing stage
- The analytical range is
relatively narrow
- Higher initial cost
- Higher running costs

Merits of furnace over flame

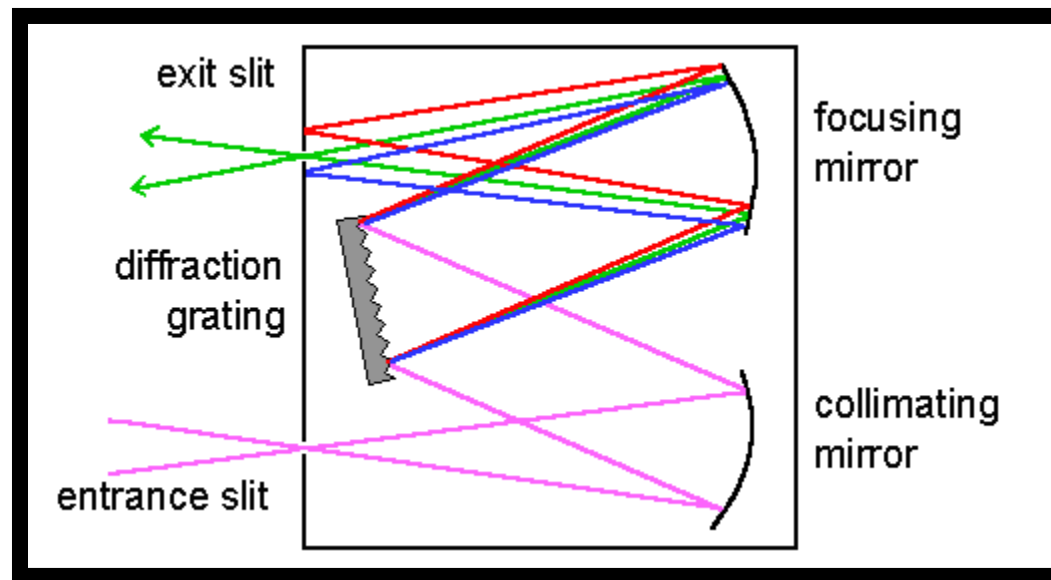
1. Residence time of the analyte in the optical path
 - ✓ Only a fraction of a second it rises via the flame.
 - ✓ Several seconds (furnace)
 - ✓ ↑ sensitivity
2. Sample volume
 - ✓ 1 – 2ml minimum for flame
 - ✓ as little as 1µl for furnace
3. Sample types
 - ✓ Liquid samples (flame)
 - ✓ A direct solid analysis without any sample preparation in addition to liquid samples analysis (furnace)

Inductively Coupled Plasma (ICP)

- ❖ Sample is aspirated into a spray chamber through a nebulizer using a system very similar to that for flame atomizer.
- ❖ However, instead of combustible gases, argon is used as a transport gas for the sample.
- ❖ The plasmas are formed by ionizing a flowing stream of argon, producing argon ions & electrons.
- ❖ The high temperatures in a plasma:
 - Result from resistive heating that develops due to the movement of the electrons and argon ions.
 - Desolvate, vaporize & largely atomize the sample.

3. Monochromator

- ❖ It selects the specific λ of light which is absorbed by the sample & transfers it to the detector, & excludes other λ .
- ❖ The selection of the specific light allows the determination of the selected element in the presence of others.



4. Detector & Read out Devices

- ❖ The light selected by the monochromator is directed onto a detector that is typically a photomultiplier tube,
 - Whose function is to convert the light signal into an electrical signal proportional to the light intensity.
- ❖ The signal could be displayed for readout , or further fed into a data station for printout by the requested format.

Effects of temperature on AS

❖ Temp determines:

- The degree to which a sample breaks down to atoms
- The extent to which a given atom is found in its ground, excited & ionized states.
 - Temp influences the strength of the signal.

❖ Boltzmann Equation: relates excited state population/ground state population ratios to energy, temperature and degeneracy.

$$\frac{N^*}{N_o} = \left(\frac{g^*}{g_o} \right) \times e^{-(E/RT)}$$

Where,

N^* = No of atoms in the excited state, N_o = No of ground state atoms,

g^*/g_o = Ratio of statistical weights for excited and ground states,

E = Energy of excitation ($= h\nu$), k = The Boltzmann's constant, T = Temp(in Kelvin)

Table 21-3 Effect of energy difference and temperature on population of excited states

Wavelength difference of states (nm)	Energy difference of states (J/atom)	Excited-state fraction (N^*/N_o) ^a	
		2 500 K	6 000 K
250	7.95×10^{-19}	1.0×10^{-10}	6.8×10^{-5}
500	3.97×10^{-19}	1.0×10^{-5}	8.3×10^{-3}
750	2.65×10^{-19}	4.6×10^{-4}	4.1×10^{-2}

a. Based on the equation $N^*/N_o = (g^*/g_o)e^{-\Delta E/kT}$ in which $g^* = g_o = 1$.

Interferences

❖ Three types of Interferences:

➤ Spectral interferences

➤ Chemical interferences

➤ Ionization interferences

Spectral Interferences

❖ Overlap of analyte signal with signals due to:

- Other elements or molecules in the sample, or
- Flame or furnace

❖ Remedy:

- Using high resolution spectrometers
 - Resolve closely spaced spectral lines
- Choose another λ for analysis

Chemical interferences

❖ Caused by any component of the sample which forms a thermally stable cpd with the analyte.

➤ ↓ extent of atomization

eg. SO_4^{2-} & PO_4^{3-} hinder the atomization of Ca^{2+}
→ forming nonvolatile salts.

Chemical interferences can be avoided/reduced by:

❖ Adding releasing agents

➤ Form thermally stable cpds with the interferents.

eg. lanthanum releases calcium from interferences like phosphate

❖ Using of protective agents

➤ eg. EDTA protects Ca^{2+} from sulfate & phosphate.

❖ Using a fuel rich flame

➤ Sufficiently decomposes a thermally stable analyte cpd

eg. nitrous oxide-acetylene flame

- High flame temp. eliminates many kinds of chemical interferences.

Ionization interferences

- ❖ Occur when the electrons are removed from the atoms which will create an ion.
- ❖ Thermal energy excites atom/removes e^- from atom.
 - Reduces the ground state atoms &
 - The absorbance reading is reduced.
 - Common with the hotter nitrous oxide-acetylene flame

Ionization interference can be eliminated by:

- ❖ Adding an excess of an element which is very easily ionized,
 - Creating a large number of free electrons in the flame
 - Suppressing the ionization of the analyte.
 - Potassium, rubidium & cesium salts - Ionization suppressants

Atomic Emission Spectroscopy

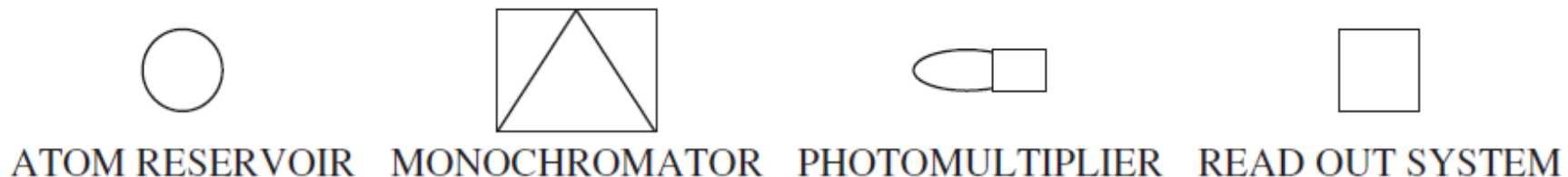
- ❖ Measures concentration by emission of light
- ❖ No light source required

- ❖ The instrumentation of AES is the same as that of AAS, but without the presence of a radiation source .

Atomic absorption spectroscopy



Atomic emission spectroscopy



Atomization & Excitation

❖ In atomic emission:

- The sample is atomized and
- The analyte atoms are excited to higher energy levels by the **same source** of thermal energy in the atomizer.

❖ The most common methods:

- Flames
- Plasmas

Flame Sources

❖ Atomization & excitation in flame atomic emission is accomplished:

➤ Using the same nebulization and spray chamber assembly used in atomic absorption.

- Flame Emission Spectroscopy

(flame AES vs flame AAS)

Plasma Sources

❖ A plasma consists of:

➤ A hot, partially ionized gas, containing an abundant conc of cations & electrons that make the plasma a conductor.

❖ The ICP torch consists of 3 concentric quartz tubes:

➤ With independent argon streams flowing through each.

➤ Surrounded at the top by a radio-frequency (RF) induction coil, which is the source of energy for the system.

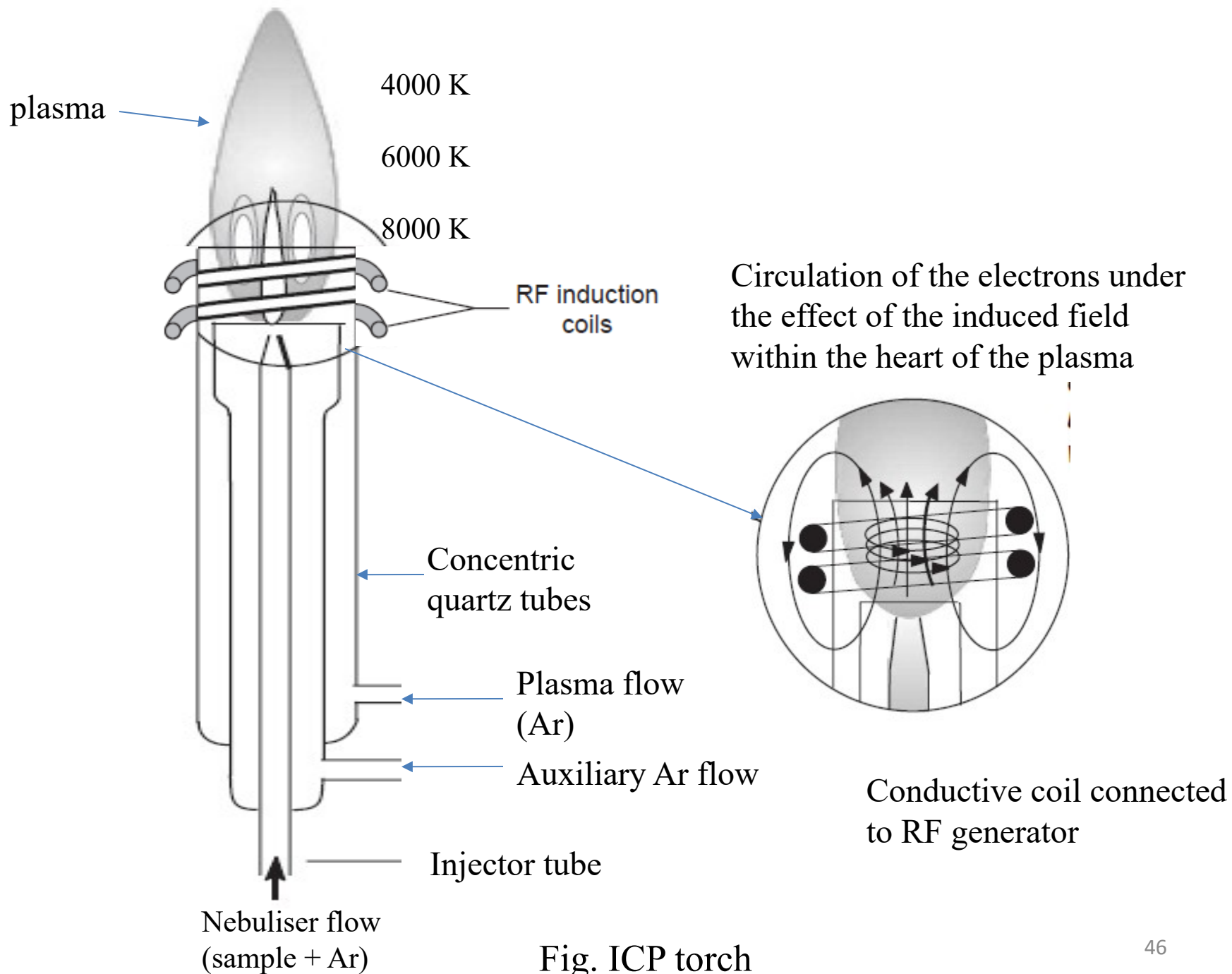
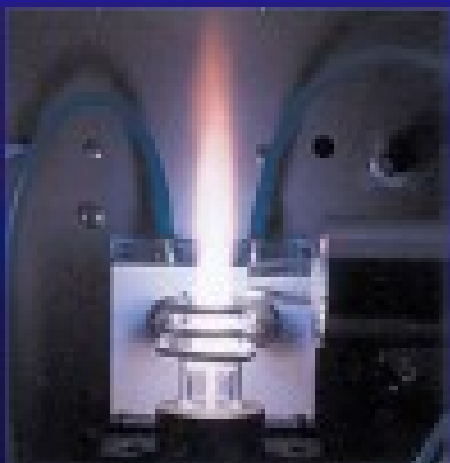
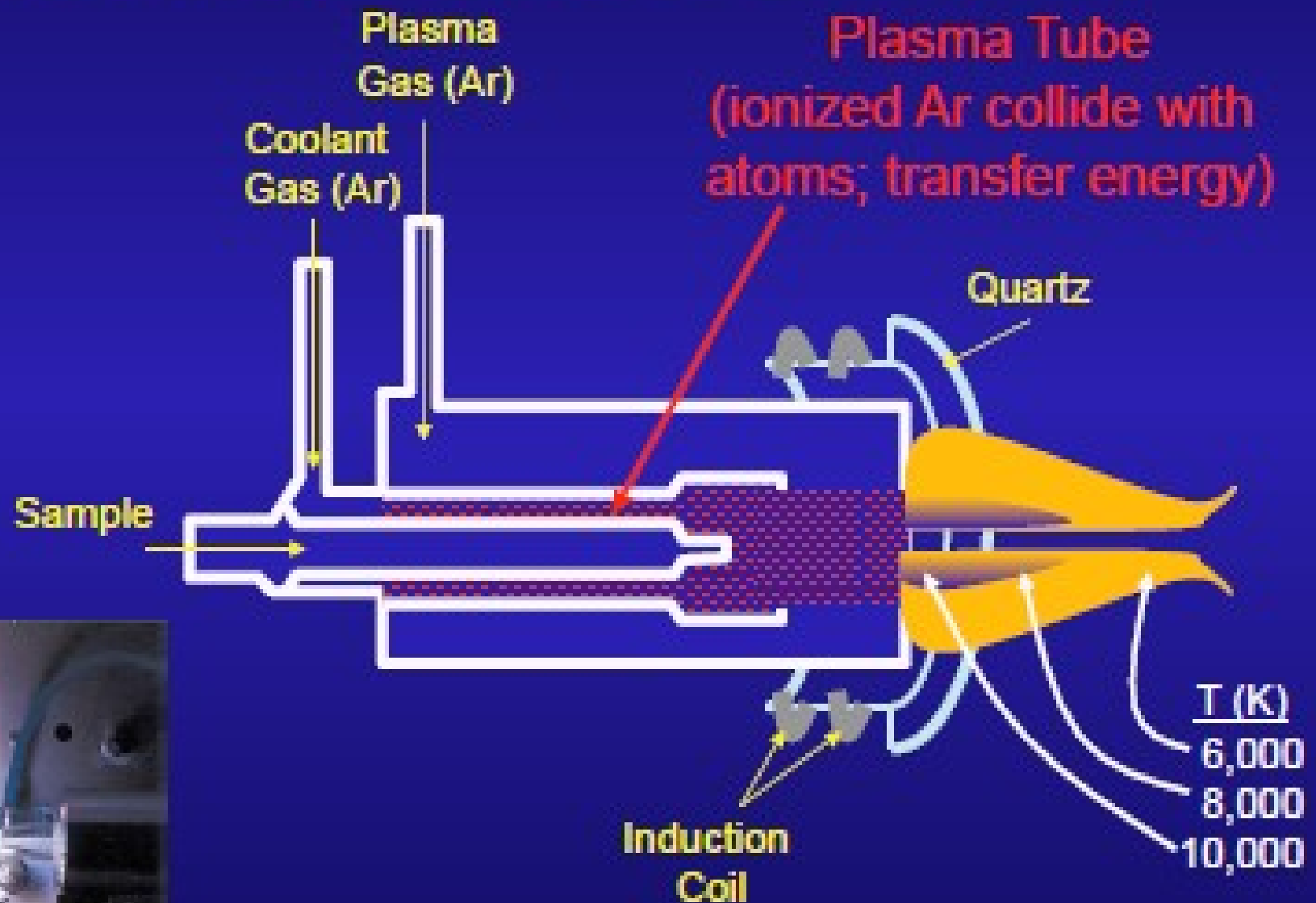


Fig. ICP torch

Inductively Coupled Plasma (ICP)



- ❖ A radiofrequency current in the induction coils
 - Creates a fluctuating magnetic field that induces the argon ions & electrons to move in a circular path.
- ❖ They are consequently accelerated, collide with argon atoms & ionize them.
- ❖ The released products by this ionization then undergo the same events.
- ❖ These colliding species cause heating of the plasma to temperatures:
 - Of about 10,000 K at the base of the plasma, and
 - Between 6000 & 8000 K at a height of 15–20 mm above the coil, where emission is usually measured.

- ❖ This temp requires thermal isolation from the outer quartz tubes
 - By introducing a high-velocity flow of argon tangentially along the walls of these tubes.
- ❖ The sample is mixed with a stream of Ar
 - Using a spray chamber nebulizer similar to that used for flame emission &
 - Carried to the hot plasma through the torch's central tube.
- ❖ The high temperatures of the plasma rapidly:
 - Desolvate, vaporize & largely **atomize** the sample.
 - Furthermore, **excite** atoms, which leads to **photon emission** & **ionization**.
 - This is why the device is found in elemental analytical methods such as inorganic mass spectrometry, ICP-MS.
- ❖ ICP-AES

❖ Advantages of plasma

- Simultaneous multi-element Analysis – saves sample amount
- Some non-metal determination (Cl, Br, I, & S)
- Plasmas operate at much higher temp than flames, they provide better atomization & more highly populated excited states.
- Excitation & emission zones are spatially separated; this results in a low background.
- Conc range of several decades ($10^5 - 10^6$)

❖ Disadvantages of plasma

- Very complex Spectra - hundreds to thousands of lines
- High resolution & expensive optical components
- Expensive instruments, highly trained personnel required

Comparison b/n AAS & AES

AAS

- Depends upon the no of ground state atoms.
- Measures the radiation absorbed by the ground state atoms.
- Presence of a light source(HCL)
- The temp in the atomizer is adjusted to atomize the analyte atoms in the ground state atoms.

AES

- Depends upon the no of excited atoms.
- Measures the radiation emitted by the excited atoms .
- Absence of the light source.
- The temp in the atomizer is big enough to atomize the analyte atoms & excite them to a higher energy level.

Application

- ❖ Quantitative analysis of elements
- ❖ Qualitative analysis of elements
 - The pattern of absorption & emission lines:
 - **Unique** for each element & can be used for identification even when several absorbing or emitting elements are present in the sample.

Application...

❖ Monitoring levels of elements(toxic/essential) in samples:

➤ Pharmaceutical products

➤ Standards

➤ Cosmetics

➤ Food supplements etc.

❖ With these information, the products can be approved or withdrawn from the markets.

Table: Assay of Pharmaceutical Substances official in BP (1993) by AAS

Name of Substance	Elements Assayed	Measured at(nm)	Limits Prescribed
Activated Charcoal	Cu, Pb, Zn	325	Cu = NMT* 25 ppm Pb = NMT 10 ppm Zn = NMT 25 ppm
Ascorbic Acid	Fe	248.3	Fe = NMT 2 ppm
Cisplatin	Ag	328	Ag =NMT 250 ppm
Copper Sulphate	Pb, Zn	Pb: 283.3 Zn: 213.9	Pb = NT 75 ppm Zn = NMT 500 ppm
Oxprenolol Hydrochloride	Pb	217	Pb = NMT 5 ppm
Prazosin Hydrochloride	Ni	232	Ni = NMT 50 ppm
Sodium Sulphite Heptahydrate	Zn	213.9	Zn = NMT 12 ppm
Zinc oxide	Cd, Pb	Cd: 228.8 Pb: 217. 0	Cd = NMT 10 ppm Pb = NMT 50 ppm

*NMT = Not More Than

Assay of Pharmaceutical Substances official in BP (1993) by Flame Ion Spectroscopy

Name of Substance	Elements Assayed	Limits Prescribed
Calcium Acetate	Mg, K, Na	NMT 500 ppm of Mg NMT 0.1% of K NMT 0.5% of Na
Magnesium Acetate	K, Na	NMT 0.1% of K NMT 0.5% of Na
Potassium Citrate	Na	NMT 0.3% of Na

End!